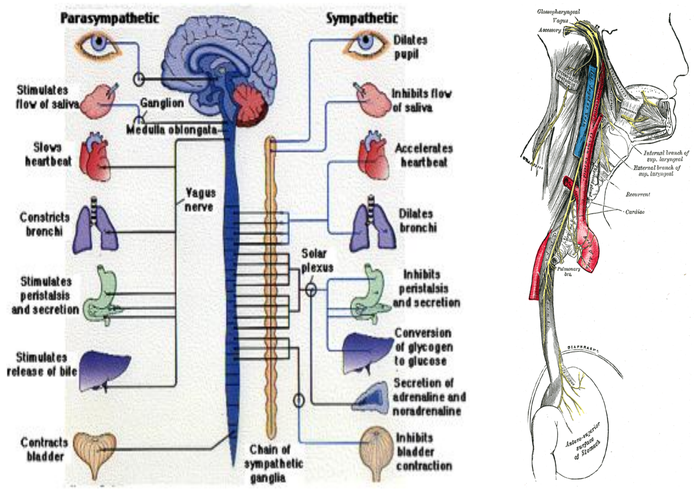
Association of Parent Vagal Tone to Emotion Regulation During Military Reintegration: Possible Gender Differences

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Introduction

This study examines the relationship between vagal tone (VT) and emotion regulation (ER) in a unique population: military parents who have recently experienced deployment (i.e. military service members and their partners/spouses). During the period of reintegration, parenting roles are strained as the recently deployed parent returns to the family system. ER is especially important for effective parenting during these times of transition. In exploration of ER during reintegration, we consider VT as a biological correlate of ER. VT is an index of autonomic flexibility (Santucci et al., 2008). Higher vagal tone is often associated with improved emotion regulation abilities (Gottman & Katz, 2002). VT is a component of the autonomic nervous system (ANS; F*igure 1*). More specifically, it is a component of the parasympathetic nervous system (PNS), which inhibits heart rate. Reducing VT readies the body for action, while increasing VT restores homeostasis. This process of activation and return to homeostasis is believed to be critical for emotion regulation. However, VT has never been examined in relation to parent ER in military parents who have recently experienced deployment.

Figure 1. How vagal tone works



Research Questions

1. What are the associations between VT and parent ER among military parent's who have recently experienced deployment ?

2. Does the relationship between VT and ER differ between mothers and fathers during reintegration?

Measures

Parents completed the Difficulties in Emotion Regulation Scale (DERS). Total ER difficulties were calculated, as well as: (a) awareness of emotions; (b) acceptance of emotions; (c) the ability to engage in goal-directed behavior; (d) the ability to refrain from impulsive behavior, when experiencing negative emotions; (e) the ability to access emotion regulation strategies perceived as effective; and (f) the ability to differentiate between emotion (i.e. clarity). Higher total scores or scores are indicative of greater problems regulating ones emotions.

Vagal tone was assessed using Polar RS800CX heart-rate monitors (*Figure 2)* during a parent-child problem solving task. VT was assessed as the ratio of low frequency to high frequency indices of heart rate variability.

All measures were completed in families' homes. All children engaged in the parent-child problem solving task were between 5 and 12 years.

Figure 2. Polar RS800CX



Participants

Forty-six parents (26 families) who recently experienced deployment were assessed. Forty-eight percent of the sample were fathers; 52% were mothers. Of the fathers, 91% were recently deployed, 9% were non-deployed spouses/partners. Of the mothers 33% were deployed; 67% were non-deployed. All parents were Caucasian; 91% were non-Hispanic. All parents had at least some college education. Most had a 4 year degree or less (91.3%); 8.7% had a graduate degree. Average household income was $80,000 to 89,000. Average number of children in household was 2.33 (.91). The results were measured in standard deviation.

Preliminary Results

Figure 1. Mean Gender Differences in VT & ER

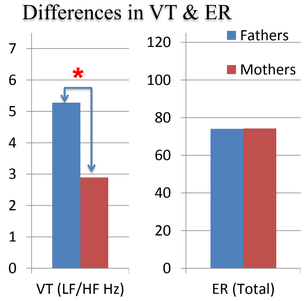


Table 1. Correlations between Demographics, VT, & ER

	VT	Education	Income	# Children
VT	NS	NS	NS	NS
Total ER	NS	-.44*	-.51*	NS
Awareness	.26+	NS	-.44*	-.39*
Acceptance	NS	-.42	-.47*	NS
Clarity	NS	-.48*	-.50*	NS
Impulse	NS	NS	-.44*	NS
Strategies	NS	-.52	NS	NS
Goals	NS	NS	-.33*	NS

*p ≤ .05 +p<.01

Does VT predict ER in Military Parents During Reintegration?

After accounting for education, income, and number of children in the home VT predicted 9% of the variance in problems with emotional awareness $R^2\Delta = .09, \beta = .32, p = .03$. VT did not predict any other measures of ER $ps \geq .17$. When controlling for gender, VT predicted awareness at the level of a nonsignificant trend $p=.09$.

Does the association between VT & ER differ by gender?

When examining the relation of vagal tone to ER separately by gender, VT did not predict any measure of ER $ps \geq .14$.

Discussion

Results suggest little support for an association between vagal tone (measured during parent-child problem solving) and emotion regulation in military parents who have recently experienced deployment. This is counter to evidence for such a relationship in other populations (Gottman & Katz 2002). Results also provided little support for an effect of gender. Lack of power due to the small sample size may have contributed to the lack of observed significant results.

Lack of power may have also contributed to failure to find significant differences in the association between VT and ER between genders. Examination of nonsignificant beta-weights in gender-specific regressions examining the association of VT to emotional awareness suggested that VT may relate to ER differently in men than women (men: β = .32, women: β = -.06).

The small sample size precluded the ability to examine the extent to which results may differ based on deployment status. The lack of racial, ethnic, and socioeconomic diversity also precluded the ability to examine the extent to which these characteristics may also impact the association between VT and ER. Results may also differ when examining VT as assessed during different tasks (e.g. emotional support or discipline tasks).

Future Directions

Future studies should include a larger sample size that is more racially, ethnically, and economically diverse. Data collection for the present study is ongoing, with efforts to enhance the diversity of the sample. Future studies also may benefit from examining, separately, deployed and non-deployed men and women because vagal tone and emotion regulation may vary depending on past or present experiences. Finally, examination of vagal tone as measured during tasks other that parent-child problems solving tasks (e.g. emotional support or discipline tasks) may enhance our understanding of any relationship between VT and ER in this unique population.

ADAPT

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